

REGULATORY DOSSIER PACKAGE

ICH CTD MODULE 3.2.S: DRUG SUBSTANCE

Major Post-Approval Amendment & Process Optimization (PAS / Type II Variation)

Submitting Entity:	Pharmaceutics Global R&D; & Regulatory Operations	Target Regulatory Authorities:	US FDA (CDER), EMA, PMDA
Drug Substance (API):	Amlodipine Besylate (USP/Ph. Eur.)	DMF Number:	US DMF 028491 / CEP 2021-104
Submission Purpose:	Post-Approval Synthetic Route Change (Step 1)	CTD Module Scope:	3.2.S.1 through 3.2.S.7 Complete Package
Document ID / Ref:	REG-AML-2026-CTD-M3	Effective Date / Year:	September 2026

1. Executive Summary & Regulatory Change Scope

This technical dossier provides a comprehensive Quality (CMC) package supporting a major post-approval amendment for Amlodipine Besylate API. The primary modifications include an optimized synthetic route for the Key Starting Material (KSM), improved crystallization kinetic controls, refined impurity qualification limits per ICH Q3A/M7, and comprehensive 24-month stability data. The updated process demonstrates complete structural comparability, reduced residual solvent burden, and enhanced batch yield consistency across commercial scale lots.

2. Master Table of Contents & Module Mapping

CTD Section	Section Name	Technical Content & Target IDP Data Scope	Page Target
3.2.S.1	General Information	Nomenclature, Structure, Physicochemical Properties (pKa, LogP, Solubility)	Page 2
3.2.S.2.2	Manufacturing Process	Amended Synthetic Scheme, Detailed Unit Operations & Operational Ranges	Page 3
3.2.S.2.3	Control of Materials	KSM Specifications, Impurity Purge Capability & Reagent Controls	Page 4
3.2.S.2.4	Process Controls (CPP/PAR)	Critical Process Parameter Matrix, Proven Acceptable Ranges & Hold Times	Page 5
3.2.S.2.6	Process Optimization (DoE)	Risk Assessments (FMEA), Continuous Crystallization Kinetics & Yields	Page 6
3.2.S.3.1	Elucidation of Structure	Spectroscopic Assignments (1H/13C-NMR, FT-IR, HRMS) & DSC/PXRD Forms	Pages 7-8
3.2.S.3.2	Impurity Profiling & Nitrosamines	ICH Q3A Impurity Limits, Purge Studies & ICH M7 Nitrosamine Risk Analysis	Pages 9-10
3.2.S.4	Control of Drug Substance	API Release Specifications, Method Validation Summary (ICH Q2 R1/R2)	Pages 11-12
3.2.S.7	Stability & Degradation	Forced Degradation (Stress Studies) & 24-Month Long-Term/Accelerated Data	Page 13
Validation	Commercial Equivalency	3-Batch Scale Commercial Analysis & GxP Regulatory Sign-Off	Page 14

3.2.S.1 General Information & Physicochemical Profile

Amlodipine Besylate is a long-acting calcium channel blocker. This section details the structural parameters, physical state, and thermodynamic properties of the active pharmaceutical ingredient (API).

Parameter	Specification / Empirical Value	Analytical Method Ref
IUPAC Chemical Name	3-ethyl 5-methyl (4RS)-4-(2-chlorophenyl)-2-[[2-(aminoethoxy)methyl]-6-methyl-1,4-dihyd ropyridine-3,5-dicarboxylate benzenesulfonate}	IUPAC Standard Rules
CAS Registry Number	111470-99-6 (Besylate Salt) / 88150-42-9 (Free Base)	Chemical Abstracts
Molecular Formula & Weight	C20H25ClN2O5 · C6H6O3S 567.05 g/mol	High-Res Mass Spectrometry
Stereochemistry	Chiral Center at C4; Marketed as Racemate (1:1 R/S mixture)	Chiral HPLC (STP-QC-109)
Melting Point / Thermal Transition	198.5°C to 201.0°C (Endothermic Peak Decomposition)	DSC / TGA (STP-QC-302)
Dissociation Constant (pKa)	8.6 at 25°C (Amine Functionality)	Potentiometric Titration
Partition Coefficient (Log P)	Log P = 3.0 (Octanol / Water System)	Shake-Flask / HPLC Method

3.2.S.1.3 Aqueous & Organic Solvent Solubility Matrix

Solubility was determined across physiological pH buffers (ICH Q6A) and standard process solvents at 25°C ± 1°C.

Solvent / Buffer Media	pH Value	Solubility (mg/mL)	USP Solubility Classification
0.1 N Hydrochloric Acid (HCl)	1.2	25.4 mg/mL	Freely Soluble
Acetate Buffer	4.5	11.2 mg/mL	Soluble
Phosphate Buffer (PBS)	6.8	1.85 mg/mL	Slightly Soluble
Purified Water	6.5	0.78 mg/mL	Very Slightly Soluble
Methanol	N/A	78.0 mg/mL	Freely Soluble
Isopropanol (IPA)	N/A	14.5 mg/mL	Sparingly Soluble
Acetone	N/A	8.2 mg/mL	Slightly Soluble

3.2.S.2.2 Manufacturing Process Flow & Narrative

The optimized manufacturing route utilizes a modified Hantzsch Ester condensation, deprotection, and benzenesulfonate salt formation. The amendment introduces a high-purity KSM route (Step 1) that suppresses impurity formation downstream.

Reaction Scheme Summary:

- Step 1 (Condensation): 2-Chlorobenzaldehyde + Ethyl 4-(2-(1,3-dioxoisindolin-2-yl)ethoxy)-3-oxobutanoate + Methyl 3-aminocrotonate → Intermediate I (Phthalimide-protected ester).
- Step 2 (Deprotection): Intermediate I + Aqueous Methylamine (40%) → Amlodipine Free Base.
- Step 3 (Salt Formation): Amlodipine Free Base + Benzenesulfonic Acid → Amlodipine Besylate (Crystalline Form I).

Unit Operations & Batch Narrative (2000 L Batch Scale)

Step	Unit Operation	Equipment Spec	Key Operating Conditions & In-Process Controls
1.1	KSM Charging & Condensation	2000 L Glass-Lined Reactor (GLR-101)	Charge 150 kg 2-chlorobenzaldehyde, 125 kg methyl 3-aminocrotonate, and 230 kg KSM in 800 L IPA. Heat to reflux (80-82°C) for 14 hrs. IPC: Unreacted aldehyde <= 0.5%.
1.2	Crystallization & Isolation	Centrifuge (CENT-101) & Filter Dryer	Cool reaction mass to 20-25°C over 3 hours. Centrifuge slurry, wash with chilled IPA (2 x 100 L). Wet intermediate cake mass: ~310 kg.
2.1	Phthalimide Deprotection	2000 L Stainless Steel (SSR-102)	Charge intermediate cake into 600 L aqueous methylamine (40%). Maintain at 42-45°C for 6 hours. IPC: Intermediate I <= 0.2% by HPLC.
2.2	Free Base Extraction	Separatory Settler Tank	Extract free base into MIBK (2 x 400 L), wash organic layer with purified water (3 x 300 L). Evaporate solvent under vacuum at <= 50°C.
3.1	Salt Formation & Refinement	2000 L Crystallizer (CRYS-103)	Dissolve free base in 600 L IPA at 50°C. Charge Benzenesulfonic Acid (1.05 eq) in IPA over 90 mins. Cool at controlled rate (0.5°C/min) to 5°C.
3.2	Drying & Milling	Cone Vacuum Dryer (CVD-101)	Dry under vacuum (>= 650 mmHg) at 45°C for 12 hours until LOD <= 0.5%. Micronize via jet mill to target PSD.

3.2.S.2.3 Control of Raw Materials & Key Starting Materials (KSM)

In alignment with ICH Q11, rigorous specifications are established for all raw materials, KSMs, reagents, and solvents to ensure downstream quality and impurity purging.

Material Name / CAS	Material Grade / Role	Acceptance Specification	Purge / Downstream Impact Rationale
Ethyl 4-(2-(1,3-dioxoisindolin-2-yl)ethoxy)-3-oxobutanoate CAS: 88150-62-3	Key Starting Material (KSM - Step 1)	Purity $\geq$ 98.5% (HPLC) Single Impurity $\leq$ 0.3% Water Content $\leq$ 0.2%	Defines core scaffold; unreacted phthalimide reagents are purged during Step 2 aqueous extractions.
2-Chlorobenzaldehyde CAS: 89-98-5	Raw Material (Step 1)	Assay $\geq$ 99.0% (GC) 2-Chlorobenzoic acid $\leq$ 0.5%	Excess aldehyde is purged in isolation slurry; acid form forms soluble salts removed in filtration.
Methyl 3-aminocrotonate CAS: 14205-39-1	Raw Material (Step 1)	Assay $\geq$ 98.0% (GC) Moisture $\leq$ 0.3%	Self-condensates are water soluble and purged in Step 2 washings.
Aqueous Methylamine CAS: 74-89-5	Reagent (Step 2)	Concentration: 39.0 - 41.0% w/w	Drives phthalimide cleavage; excess amine is volatile and purged in vacuum drying.
Benzenesulfonic Acid CAS: 98-11-3	Reagent (Step 3)	Assay $\geq$ 98.0% Free H ₂ SO ₄ $\leq$ 0.3%	Forms target salt; controlled stoichiometry prevents free base carryover.
Isopropanol (IPA) CAS: 67-63-0	Process Solvent (Class 3)	Purity $\geq$ 99.8% Non-volatile matter $\leq$ 0.001%	Primary reaction/crystallization media; regulated per ICH Q3C.

3.2.S.2.4 Process Control Matrix (CPP / PAR) & Hold Times

Critical Process Parameters (CPPs) and Proven Acceptable Ranges (PARs) were established via Quality by Design (QbD) DoE studies to guarantee consistent API quality.

Process Step	Parameter (CPP vs Non-CPP)	Target Parameter	Proven Acceptable Range (PAR)	Impact on Critical Quality Attribute (CQA)
Step 1: Condensation	Reaction Temp (CPP)	81.0°C	78.0°C - 84.0°C	Temperature > 85°C increases oxidative pyridine impurity A.
Step 1: Condensation	Agitation Speed (Non-CPP)	85 rpm	70 - 100 rpm	Ensures thermal mass uniformity; minimal effect on profile.
Step 2: Deprotection	Methylamine Addition Rate (CPP)	2.0 L/min	1.5 - 2.5 L/min	Exotherm control; prevents degradation of dimer impurity.
Step 2: Deprotection	Reaction Time (CPP)	6.0 Hours	5.0 - 8.0 Hours	Time < 5 hrs leaves incomplete deprotection (>0.2% Int I).
Step 3: Salt Formation	Acid Addition Duration (CPP)	90 Minutes	75 - 120 Minutes	Rapid addition causes amorphous salt precipitation.
Step 3: Crystallization	Cooling Rate (CPP)	0.5°C / min	0.3°C - 0.7°C / min	Controls particle size distribution (D90) and polymorphic Form I.
Step 3: Vacuum Drying	Drying Temp / Pressure (CPP)	45°C / >=650 mmHg	40°C - 50°C	Excess heat causes degradation; controls residual IPA < 3000 ppm.

Validated In-Process Intermediate Hold-Time Limits

Intermediate Stage	Storage Condition	Validated Maximum Hold Time	Quality Control Re-test Parameter
Step 1 Wet Cake (Int I)	Ambient (20-25°C), Sealed Drum	7 Days	Related Substances & Purity (HPLC)
Step 2 Free Base Solution	Stainless Tank at 15-20°C	48 Hours	Assay & Color Variation Index
Step 3 Crude Wet Besylate	Chilled Stainless Vessel (5-10°C)	72 Hours	Polymorphic Form Integrity (PXRD)
Final Micronized API	Double PE Bag in HDPE Drum	36 Months	Full Release Specification Retest

3.2.S.2.6 Process Development History & Optimization (DoE)

A Design of Experiments (DoE) response surface methodology was deployed to optimize Step 3 crystallization kinetics. The objective was maximizing yield while minimizing particle size variance and impurity entrainment.

Exp Run	Cooling Rate (°C/min)	Seed Temperature (°C)	Agitation (RPM)	Yield (%)	PSD D90 (µm)	Impurity A (%)
DoE-01	0.2	45	60	82.1%	125 µm	0.04%
DoE-02	1.0	45	60	88.5%	35 µm	0.12%
DoE-03	0.2	55	100	83.0%	110 µm	0.03%
DoE-04	1.0	55	100	89.2%	28 µm	0.15%
DoE-05 (Center)	0.5	50	80	88.4%	62 µm	0.05%
DoE-06 (Center)	0.5	50	80	88.6%	60 µm	0.04%
DoE-07 (Optimal)	0.5	50	80	88.8%	58 µm	0.04%

FMEA Risk Assessment Summary (Pre vs Post Optimization)

Failure Mode / Risk Area	Initial Severity (S)	Initial Occurrence (O)	Initial RPN	Corrective Optimization / Action	Final RPN
Overheating Step 1 → Impurity A	4	3	36 (Medium)	Tightened PAR to 78-84°C with automated DCS interlocks.	12 (Low)
Uncontrolled Salt Crystallization	4	4	48 (High)	Implemented controlled 0.5°C/min ramp rate and seeding.	12 (Low)
Excess Residual IPA (>3000 ppm)	3	3	27 (Medium)	Optimized cone vacuum dryer temperature profile & time.	9 (Low)

3.2.S.3.1 Structural Elucidation — Spectroscopy Data

Chemical structure confirmation of Amlodipine Besylate was performed via 1H-NMR, 13C-NMR, FT-IR, and High-Resolution Mass Spectrometry (HRMS).

Chemical Shift (δ, ppm)	Multiplicity	Integration (#H)	Structural Assignment (Amlodipine Core)
1.18	t (triplet)	3H	-OCH2CH3 (Ester Ethyl Group)
2.32	s (singlet)	3H	-CH3 (Pyridine Ring C6-CH3)
3.58	s (singlet)	3H	-OCH3 (Ester Methyl Group)
4.02	m (multiplet)	2H	-CH2-O-CH2-CH2-NH2
4.70	s (singlet)	2H	-O-CH2-Pyridine
5.40	s (singlet)	1H	Chiral C4-H (Dihydropyridine)
7.10 - 7.35	m (multiplet)	4H	2-Chlorophenyl Ring Protons
7.30 - 7.60	m (multiplet)	5H	Benzenesulfonate An Anion Protons

FT-IR & Mass Spectrometry Summary

Technique	Observed Value	Theoretical / Reference	Conclusion / Assignment
FT-IR Wavenumber	1685 cm ⁻¹	1680 - 1690 cm ⁻¹	C=O Ester Conjugated Stretch
FT-IR Wavenumber	1210 cm ⁻¹ & 1042 cm ⁻¹	1200 & 1050 cm ⁻¹	Sulfonate Salt (S=O) Strong Bands
HRMS (ESI+)	m/z 409.1528 [M+H] ⁺	409.1530 Da	Confirms Amlodipine Free Base C20H26ClN2O5
Elemental Analysis	C: 54.95%, H: 5.48%, N: 4.90%	C: 55.02%, H: 5.51%	Conforms to C26H31ClN2O8S Monosalt

3.2.S.3.1 Solid State & Polymorphic Form Characterization

Powder X-Ray Diffraction (PXRD) and Differential Scanning Calorimetry (DSC) confirm that the optimized process consistently yields Anhydrous Form I.

Diffraction Angle (2θ ± 0.2°)	Relative Intensity (%)	Interplanar Spacing d(Å)	Polymorphic Form Characterization
9.8°	100%	9.01 Å	Characteristic Form I Peak (Dominant)
11.6°	45%	7.62 Å	Form I Specific Reflection
12.2°	38%	7.25 Å	Form I Specific Reflection
16.5°	62%	5.37 Å	Form I Core Lattice Plane
22.8°	78%	3.90 Å	Form I High-Intensity Cluster
24.1°	30%	3.69 Å	Form I Secondary Reflection

Thermal Analysis (DSC / TGA Summary)

Parameter	Thermal Result	Acceptance Criterion	Interpretation
DSC Endothermic Peak	199.8°C (Onset: 198.2°C)	198.0°C - 202.0°C	Sharp melting endotherm of Form I
Enthalpy of Fusion (ΔHf)	89.2 J/g	85.0 - 95.0 J/g	High crystalline purity
TGA Mass Loss (25-150°C)	0.18% w/w	<= 0.5% w/w	Confirms Anhydrous (Non-solvated) Form

3.2.S.3.2 Impurity Profiling & ICH Thresholds

All process and degradation impurities were qualified according to ICH Q3A(R2). Limits are assigned based on maximum daily dose (10 mg/day).

Impurity Name / Structure ID	Origin / Mechanism	ICH Q3A Limit	Proposed Spec	Batch Range (n=10)
Impurity A: Oxidative Pyridine analogue CAS: 113994-41-5	Process / Degradant (Oxidation)	0.15%	NMT 0.10%	< 0.02% - 0.04%
Impurity B: Methyl ester analogue CAS: 103863-88-9	Raw Material Transesterification	0.15%	NMT 0.10%	< 0.01% - 0.03%
Impurity D: Phthalimide intermediate CAS: 88150-62-3	Step 2 Incomplete Reaction	0.15%	NMT 0.10%	ND (< 0.01%)
Impurity F: Methyl ethyl ester analogue CAS: 140171-66-0	Side Condensation Step 1	0.15%	NMT 0.15%	0.03% - 0.06%
Any Unspecified Impurity	Unknown / Synthetic	0.10%	NMT 0.05%	< 0.03%
Total Impurities	Cumulative Profile	N/A	NMT 0.50%	0.08% - 0.14%

Residual Solvents Profile (ICH Q3C)

Solvent Name	ICH Class	ICH Q3C Option 1 Limit	Proposed QC Spec	Observed Level
Isopropanol (IPA)	Class 3	5000 ppm	NMT 3000 ppm	450 - 820 ppm
Methyl Isobutyl Ketone (MIBK)	Class 2	450 ppm	NMT 250 ppm	12 - 35 ppm
Methanol	Class 2	3000 ppm	NMT 500 ppm	ND (< 10 ppm)

3.2.S.3.2 Nitrosamine Risk Assessment & Purge Evaluation

In compliance with US FDA and EMA Guidance on Nitrosamine Impurities, an evaluation was conducted for potential N-nitrosamine formation (ICH M7).

Evaluation Factor	Risk Assessment & Chemical Pathway Finding	Conclusion / Action Level
Secondary / Tertiary Amines	Amlodipine contains a primary aliphatic amine; secondary amine impurities are monitored below 0.05%.	LOW RISK: No precursors present.
Nitrosating Agents	Sodium Nitrite or Nitrates are NOT utilized in any step of the synthetic process.	NO RISK: Absence of nitrites.
Solvent Recycling & Equipment	Freshly distilled solvents are used; zero cross-contamination risk from external recovered lines.	NO RISK: Dedicated equipment.
NDSRI Risk Assessment	N-Nitroso-Amlodipine potential studied. Ames test confirms negative mutagenicity.	Calculated AI: 1500 ng/day.
Confirmatory LC-MS/MS Testing	Testing of 3 commercial scale lots for 8 major nitrosamines (NDMA, NDEA, NDIPA, NDBA, etc.).	ALL ND (< 0.03 ppm / LOQ).

Elemental Impurities Summary (ICH Q3D)

Element Class	Target Elements	ICH Q3D Daily PDE (µg/day)	Observed Level (ICP-MS)
Class 1	Cd, Pb, As, Hg	15, 5, 15, 30 µg/day	< 10% of PDE (Conforms)
Class 2A	V, Ni, Co	100, 200, 50 µg/day	< 10% of PDE (Conforms)
Class 3	Cu, Cr, Mo	3000, 11000, 3000 µg/day	< 5% of PDE (Conforms)

3.2.S.4 Finished API Specification Sheet

Release specifications for Amlodipine Besylate API meet USP, Ph. Eur., and ICH Q6A standards.

Test Parameter	Test Method ID	Acceptance Specification / Release Threshold
Appearance	Visual (STP-QC-101)	White or almost white crystalline powder
Identification (FT-IR)	STP-QC-102	Conforms to USP Reference Standard
Identification (Besylate Salt)	STP-QC-103	Positive test for Benzenesulfonate
Assay (HPLC)	STP-QC-401	98.0% - 102.0% (Dried Basis)
Organic Impurities	STP-QC-402	Imp A <= 0.10%, Imp B <= 0.10%, Total <= 0.50%
Residual Isopropanol	STP-GC-105	NMT 3000 ppm
Water Content (KF)	STP-QC-205	NMT 0.5% w/w
Residue on Ignition	STP-QC-208	NMT 0.1% w/w
Particle Size Distribution	STP-QC-305 (Laser)	D10: 5-15 µm, D50: 25-40 µm, D90 <= 80 µm
Bacterial Endotoxins	STP-QC-501 (LAL)	NMT 0.35 EU/mg (Parenteral Grade)

3.2.S.4.3 Analytical Method Validation Summary (ICH Q2 R1/R2)

Validation metrics for the HPLC Assay and Related Substances procedure (STP-QC-401/402) are summarized below.

Validation Parameter	Acceptance Criteria	Assay (STP-QC-401)	Related Substances (STP-QC-402)
Specificity / Selectivity	No interference at Amlodipine peak	No co-eluting peaks	Peak Purity Angle < Threshold
Linearity Range	R ² >= 0.999	R ² = 0.9998 (50-150% Target)	R ² = 0.9995 (0.05-1.5% Range)
Accuracy (% Recovery)	98.0% - 102.0% (Assay) / 85-115% (Imp)	99.4% - 100.8%	94.2% - 103.5%
Repeatability (%RSD)	%RSD <= 1.0% (Assay) / <= 5.0% (Imp)	0.35% (n=6)	1.8% (n=6)
Intermediate Precision	%RSD <= 1.5% across 2 analysts/days	0.52% overall	2.4% overall
LOD & LOQ Thresholds	LOQ <= 0.05% for impurities	N/A	LOD: 0.01%, LOQ: 0.03%
Solution Stability	Stable for >= 48 hours at 2-8°C	Stable 72 Hours	Stable 48 Hours

3.2.S.7 Forced Degradation & 24-Month Stability Data

Stress testing (forced degradation) demonstrates the stability-indicating capability of the analytical method. Real-time and accelerated stability studies support a 36-month retest period.

Stress Condition	Exposure Parameters	Assay (%)	Total Degradants (%)	Primary Degradation Product
Acid Hydrolysis	1 N HCl, 60°C, 6 Hours	91.2%	8.2%	Impurity D (Phthalimide Base)
Base Hydrolysis	1 N NaOH, 60°C, 6 Hours	94.5%	5.1%	Ester Hydrolysis Cleavage
Oxidative Stress	3% H2O2, Room Temp, 2 Hours	88.4%	11.2%	Impurity A (Oxidative Pyridine)
Thermal Stress	105°C, Dry Heat, 48 Hours	98.1%	1.6%	Minor Unspecified Degradants
Photolytic Stress	1.2 Million Lux Hours (ICH Q1B)	92.0%	7.5%	Photo-cyclized Pyridine Derivative

24-Month Long-Term Stability Data Summary (25°C ± 2°C / 60% ± 5% RH)

Test Parameter	Initial (T0)	6 Months	12 Months	18 Months	24 Months	Specification
Appearance	White Powder	Conforms	Conforms	Conforms	Conforms	White Crystalline Powder
Assay (%)	99.8%	99.7%	99.5%	99.4%	99.2%	98.0% - 102.0%
Impurity A (%)	0.02%	0.03%	0.03%	0.04%	0.04%	NMT 0.10%
Total Impurities	0.08%	0.10%	0.11%	0.13%	0.14%	NMT 0.50%
Water Content (%)	0.21%	0.22%	0.24%	0.25%	0.26%	NMT 0.50%

Commercial Batch Analysis & Regulatory Sign-Off

Comparative release analytical data across 3 consecutive commercial scale validation lots (200 kg) manufactured using the amended Step 1 process.

Analytical Parameter	Lot # AML-2026-01	Lot # AML-2026-02	Lot # AML-2026-03	Target Specification
Batch Yield (kg / %)	177.2 kg (88.6%)	176.8 kg (88.4%)	177.6 kg (88.8%)	Expected: 85.0% - 90.0%
Assay (HPLC Basis)	99.8%	99.6%	99.9%	98.0% - 102.0%
Impurity A	0.02%	0.03%	0.02%	NMT 0.10%
Impurity F	0.04%	0.05%	0.04%	NMT 0.15%
Residual IPA	520 ppm	480 ppm	610 ppm	NMT 3000 ppm
Polymorphic Form	Form I (Anhydrous)	Form I (Anhydrous)	Form I (Anhydrous)	Conforms to Form I
PSD D50 (Micronized)	32.4 µm	31.8 µm	33.1 µm	25.0 - 40.0 µm

Regulatory Handover & Quality Authorization

This technical dossier has been prepared in compliance with GxP guidelines and reviewed by cross-functional technical heads. The data provided herein accurately reflects the manufacturing process and quality controls.

Function Role	Author / Approver Name	Signature	Date Stamp
Lead Process Chemist	Dr. A. R. Vance, PhD	[Electronically Signed]	September 1, 2026
Head of Analytical R&D;	Dr. E. M. Thorne, PhD	[Electronically Signed]	September 1, 2026
VP, Quality Assurance	M. S. Jenkins, MS, CQA	[Electronically Signed]	September 1, 2026
Regulatory Affairs Director	S. K. Patel, RAC	[Electronically Signed]	September 1, 2026